

Data Mining Methodologies

SEMMA / Enterprise Miner

(Sample, Explore, Modify, Model and Assess)

SAS Institute

CRISP-DM / Clementine / SPSS / IBM ...

Consortium NCR + DaimlerChrysler + SPSS + OHRA
(excluded in version 1.00)

CRISP-DM

Cross-Industry Standard Process for Data Mining

CRISP-DM Hierarchical breakdown

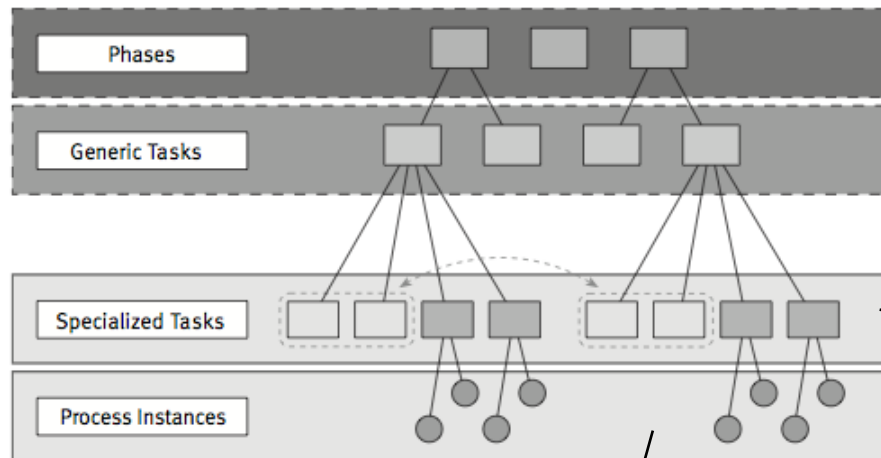


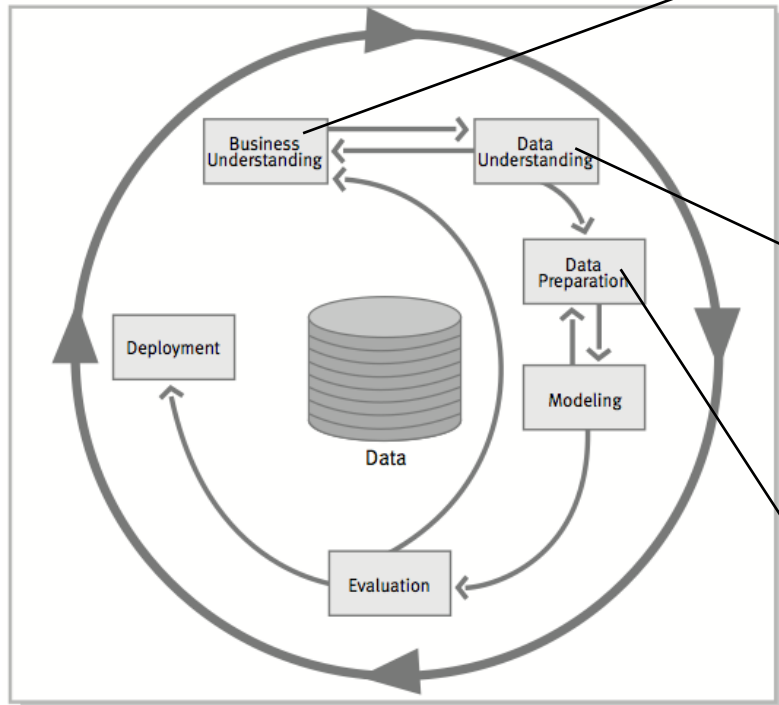
Figure 1: Four level breakdown of the CRISP-DM methodology

The fourth level is a record of the actions, decisions, and results of an actual data mining engagement

- CRISP Process Model
- Mapping
- CRISP Process
- At the top level, the data mining process is organized into a number of phases
 - each phase consists of several second-level generic tasks
 - describe how actions should be carried out in certain specific situations

For example, at the second level there might be a generic task called clean data. The third level describes how this task differs in different situations, such as cleaning numeric values versus cleaning categorical values, or whether the problem type is clustering or predictive modeling

CRISP-DM Reference Model



understanding the project objectives and requirements from a business perspective, then converting this knowledge into a data mining problem definition and a preliminary plan designed to achieve the objectives

starts with initial data collection and proceeds with activities that enable you to become familiar with the data, identify data quality problems, discover first insights into the data, and/or detect interesting subsets to form hypotheses regarding hidden information

covers all activities needed to construct the final dataset [data that will be fed into the modeling tool(s)] from the initial raw data.

Tasks include table, record, and attribute selection, as well as transformation and cleaning of data for modeling tools

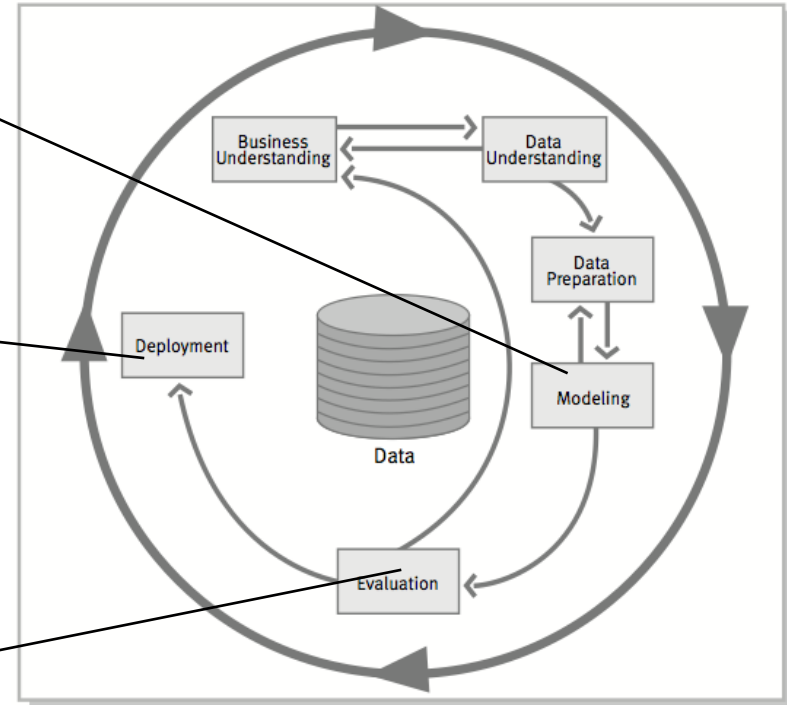
CRISP-DM Reference Model

various modeling techniques are selected and applied, and their parameters are calibrated to optimal values

deployment phase can be as simple as generating a report or as complex as implementing a repeatable data mining process across the

evaluate the models and review the steps executed to create it, to be certain the model properly achieves the business objectives

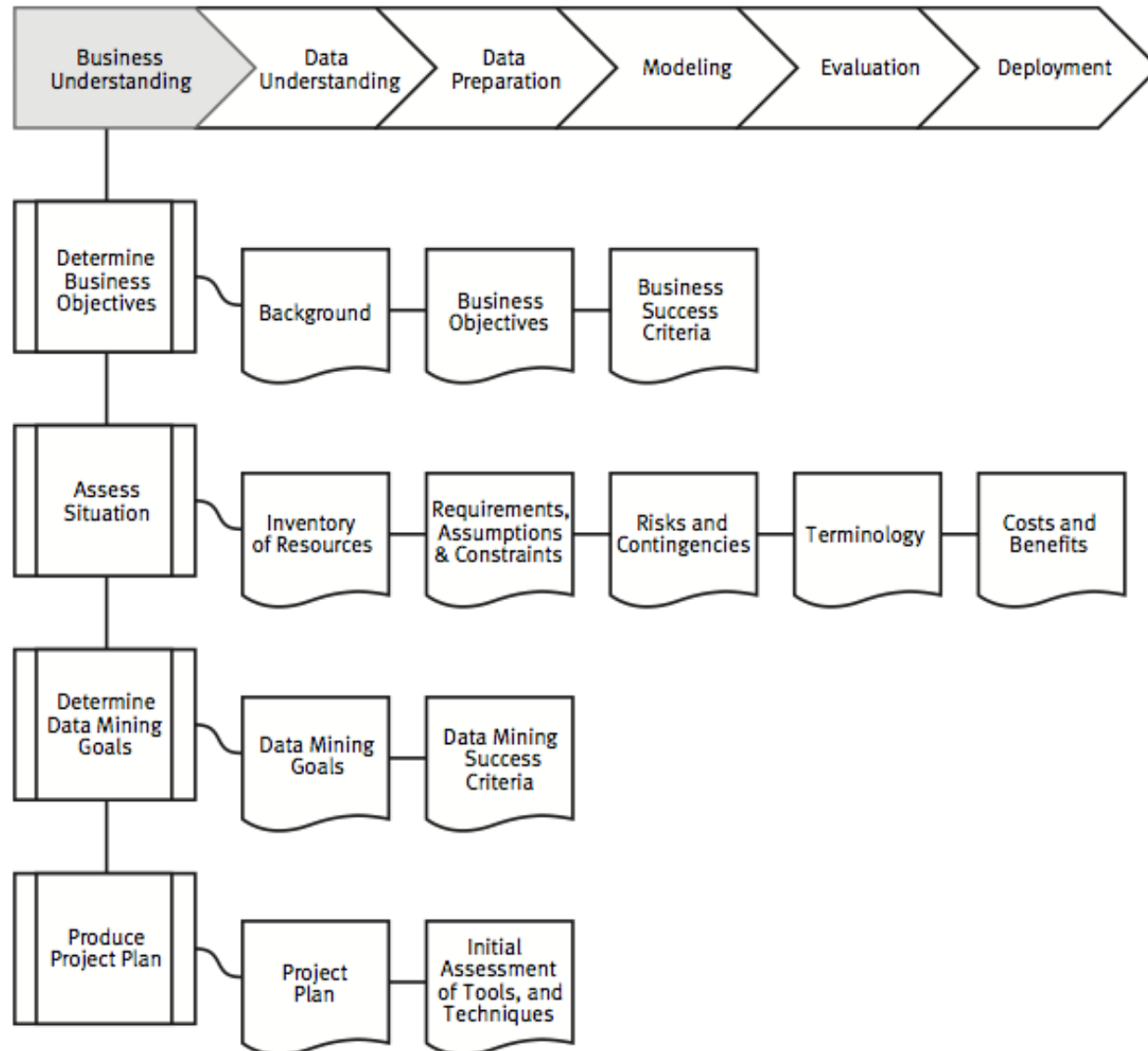
A key objective is to determine if there is some important business issue that has not been sufficiently considered. At the end of this phase, a decision on the use of the data mining results should be reached



Main Phases: Tasks and *Outputs*

Business Understanding	Data Understanding	Data Preparation	Modeling	Evaluation	Deployment
Determine Business Objectives <i>Background</i> <i>Business Objectives</i> <i>Business Success Criteria</i>	Collect Initial Data <i>Initial Data Collection Report</i>	Select Data <i>Rationale for Inclusion/Exclusion</i>	Select Modeling Techniques <i>Modeling Technique</i> <i>Modeling Assumptions</i>	Evaluate Results <i>Assessment of Data Mining Results w.r.t. Business Success Criteria</i> <i>Approved Models</i>	Plan Deployment <i>Deployment Plan</i>
Assess Situation <i>Inventory of Resources</i> <i>Requirements, Assumptions, and Constraints</i> <i>Risks and Contingencies</i> <i>Terminology</i> <i>Costs and Benefits</i>	Describe Data <i>Data Description Report</i>	Clean Data <i>Data Cleaning Report</i>	Generate Test Design <i>Test Design</i>	Review Process <i>Review of Process</i>	Plan Monitoring and Maintenance <i>Monitoring and Maintenance Plan</i>
Determine Data Mining Goals <i>Data Mining Goals</i> <i>Data Mining Success Criteria</i>	Explore Data <i>Data Exploration Report</i>	Construct Data <i>Derived Attributes</i> <i>Generated Records</i>	Build Model <i>Parameter Settings</i> <i>Models</i> <i>Model Descriptions</i>	Determine Next Steps <i>List of Possible Actions</i> <i>Decision</i>	Produce Final Report <i>Final Report</i> <i>Final Presentation</i>
Produce Project Plan <i>Project Plan</i> <i>Initial Assessment of Tools and Techniques</i>	Verify Data Quality <i>Data Quality Report</i>	Integrate Data <i>Merged Data</i>	Assess Model <i>Model Assessment</i> <i>Revised Parameter Settings</i>		Review Project <i>Experience</i> <i>Documentation</i>
		Format Data <i>Reformatted Data</i>			
		<i>Dataset</i> <i>Dataset Description</i>			

Business Understanding



Background (a kind of STAR)

“DM approaches demonstrated to be useful improving mobile communication business”...

Business objectives and success criteria

Retain customers decreasing the churn rate (customers moving out < 5%) (Utility)

Inventory of resources

1 Business expert, 3 Data base experts, DB Server, DM Server, Data Sets, Budget...

Requirements, assumptions, and constraints

The DM models should be readable, legal aspects

Risks and contingencies

Financial support depends on the results and is committed with the actual CIO

Terminology

Glossary on DM, glossary on mobile communications business and operation

Costs and benefits

50 K €. Revenue of 100 € for each customer not moved => retain more than 500 customers

Data Mining Objectives

Classification Task

Classification Model to determine whether a customer will move out or not

Data Mining Success Criteria

90 % of minimum accuracy (validity)

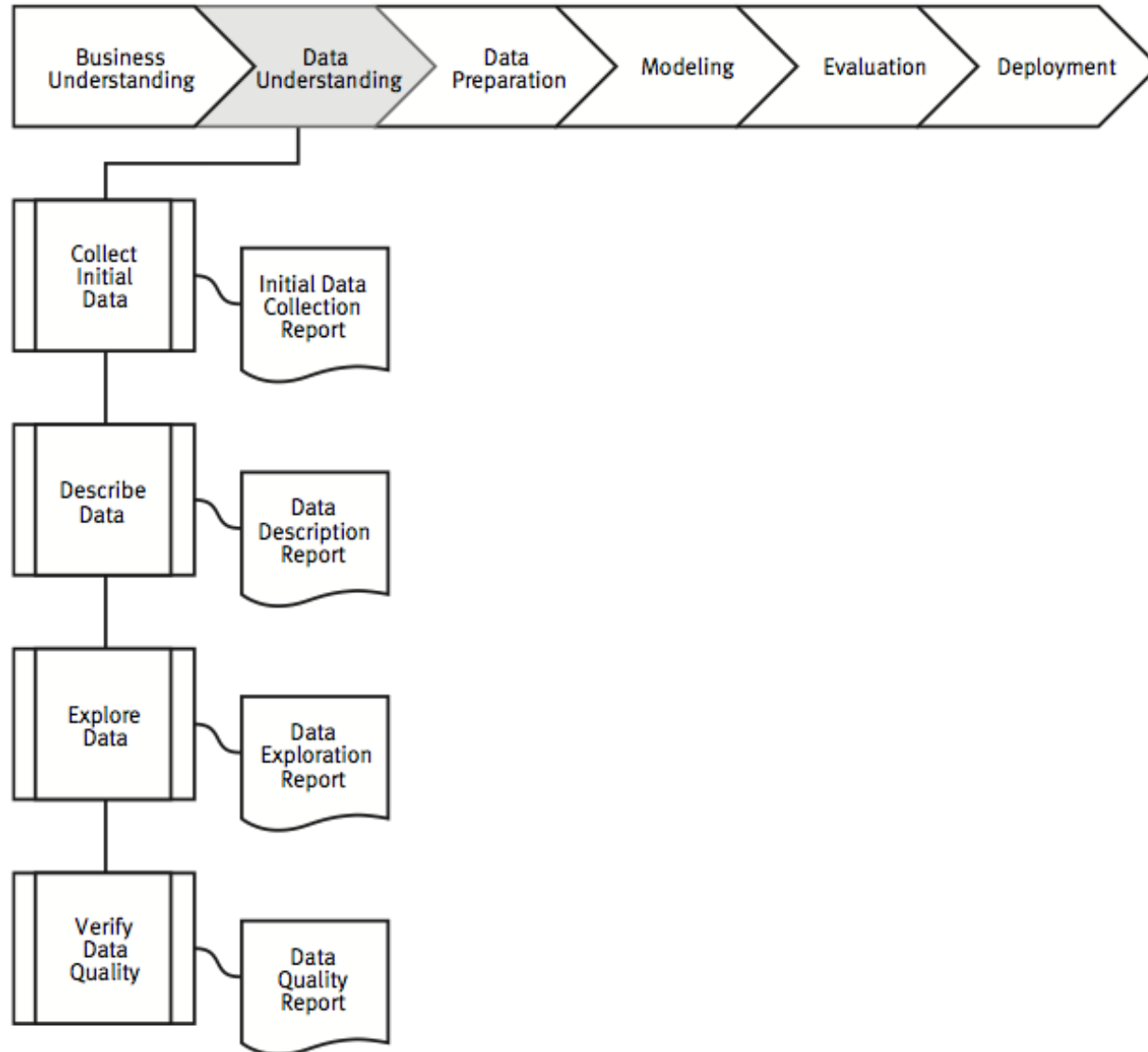
Project plan

Phases, milestones, gantt diagram, resources, ...

Initial assessment of tools and techniques

Clementine and Weka available and installed in the computers

Data Understanding



Initial data collection report

- Background of data
- List of data sources
- Problems encountered in data acquisition or extraction

Data description report

Each data source described in detail

- List of tables (may be only one) or other database objects
- Description of each field

(data dictionary)

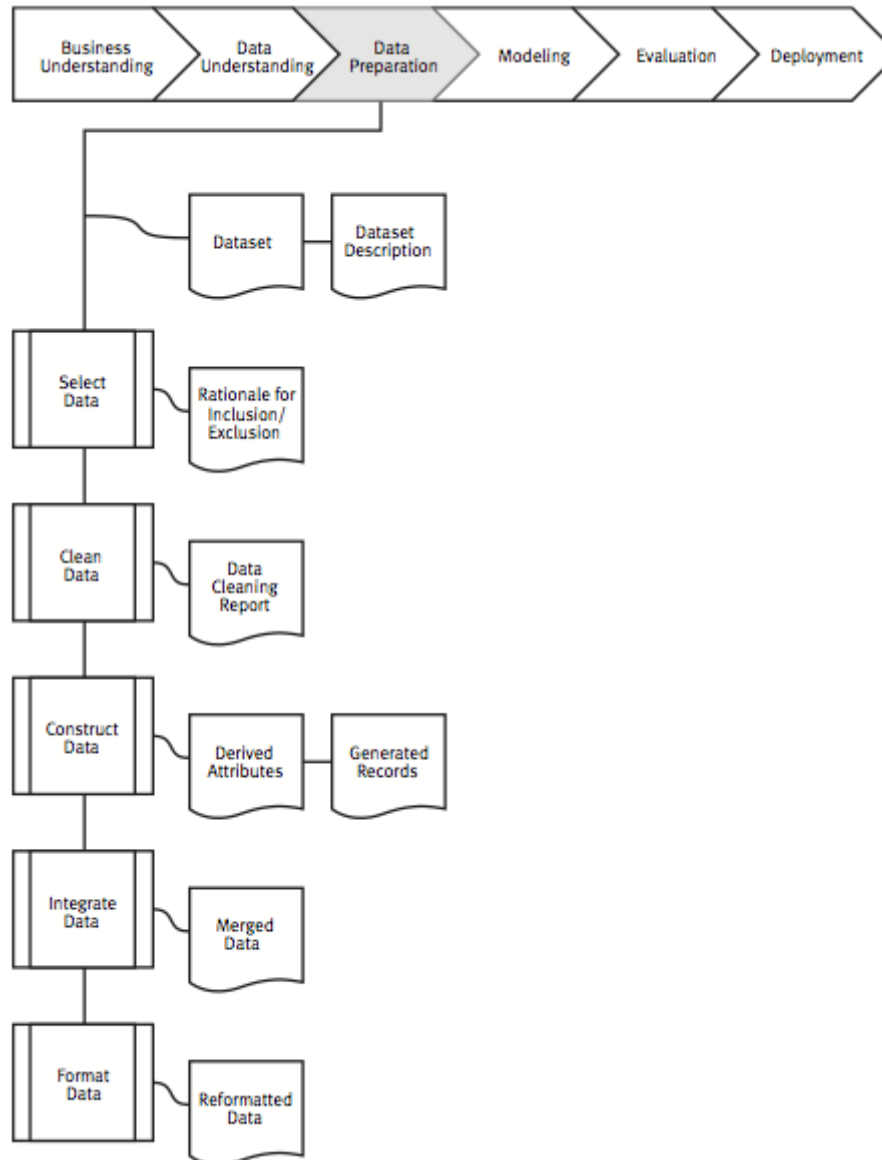
Data exploration report

- Expected regularities or patterns
- Method of detection
- Regularities or patterns found, expected and unexpected
- Any other surprises
- Conclusions for data transformation, data cleaning, and any other pre-processing
- Conclusions related to data mining goals or business objectives
- Summary of conclusions

Data quality report

- Approach taken to assess data quality
- Results of data quality assessment
- Summary of data quality conclusions

Data Preparation



Dataset description report

This report provides a description of the dataset (after pre-processing)

- Background, including broad goals and plan for pre-processing
- Rationale for inclusion/exclusion of datasets.

For each included dataset

- Description of the pre-processing, including the actions that were necessary to address any data quality issues
- Detailed description of the resulting dataset, table by table and field by field
- Rationale for inclusion/exclusion of attributes
- Discoveries made during pre-processing, and any implications for further work– Summary and conclusions

Modeling

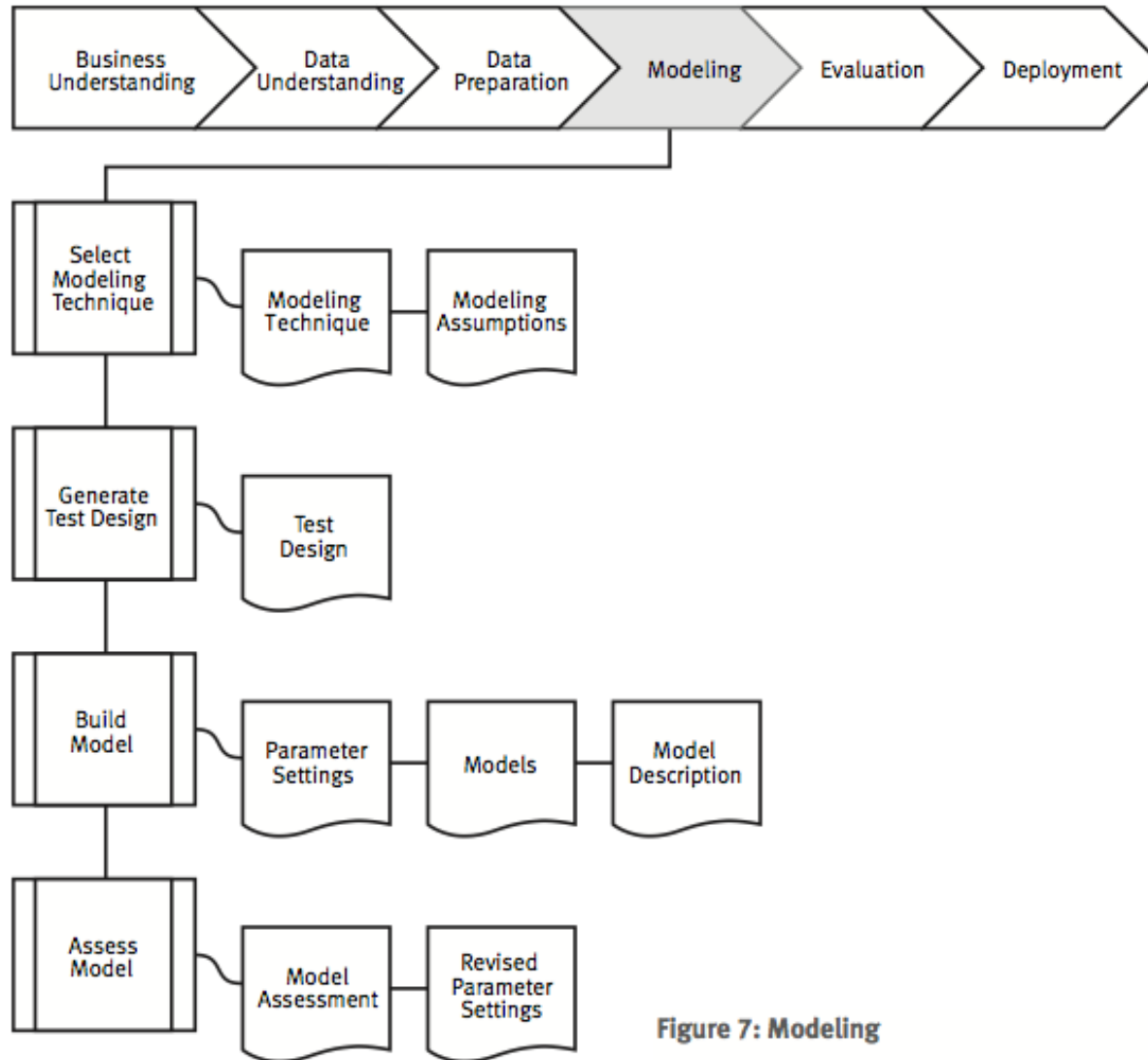


Figure 7: Modeling

•Test design

For each modeling task:

- Broad description of the type of model and the training data to be used
- Explanation of how the model will be tested or assessed
- Description of any data required for testing
- Plan for production of test data if any
- Description of any planned examination of models by domain or data experts
- Summary of test plan

•Model description

For each model:

- Type of model and relationship to datamining goals
- Parameter settings used to produce the model
- Detailed description of the model and any special features

For example:

For rule-based models, list the rules produced plus any assessment of per-rule or overall model accuracy and coverage

For opaque models, list any technical information about the model (such as neural network topology) and any behavioral descriptions produced by the modeling process (such as accuracy or sensitivity)

- Description of the model's behavior and interpretation
- Conclusions regarding patterns in the data (if any).

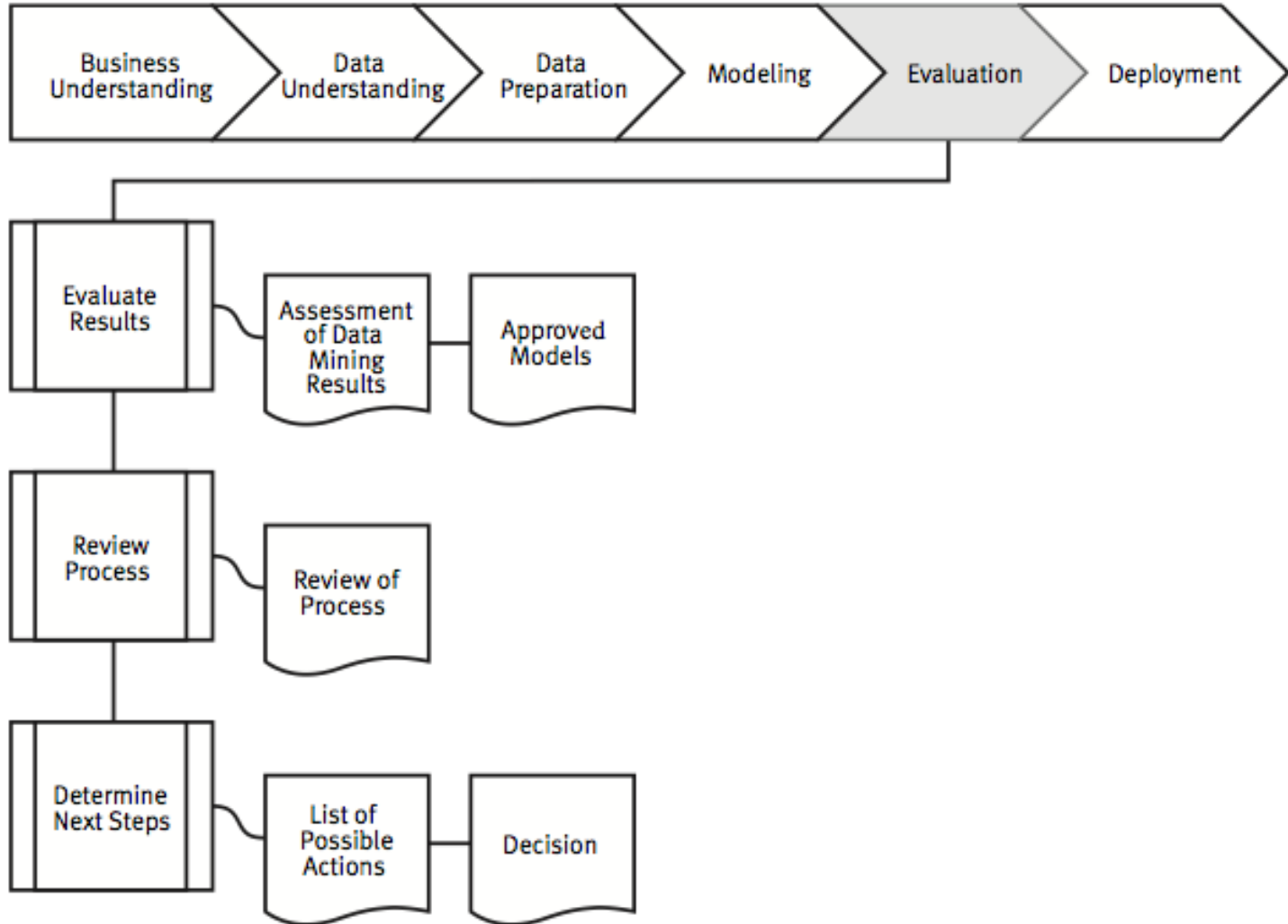
Model assessment

Overview of assessment process and results, including any deviations from the test plan

For each model:

- Detailed assessment, including measurements such as accuracy and interpretation of behavior (ROC, AUC,...)
- Any comments on models by domain or data experts
- Summary assessment of models
- Insights into why a certain modeling technique and certain parameter settings led to good/bad results
- Summary assessment

Evaluation



Review of business objectives and business success criteria (which may have changed during and/or as a result of data mining).

For each business success criterion:

- Detailed comparison between success criterion and datamining results
- Conclusions about achievability of success criterion and suitability of datamining processs

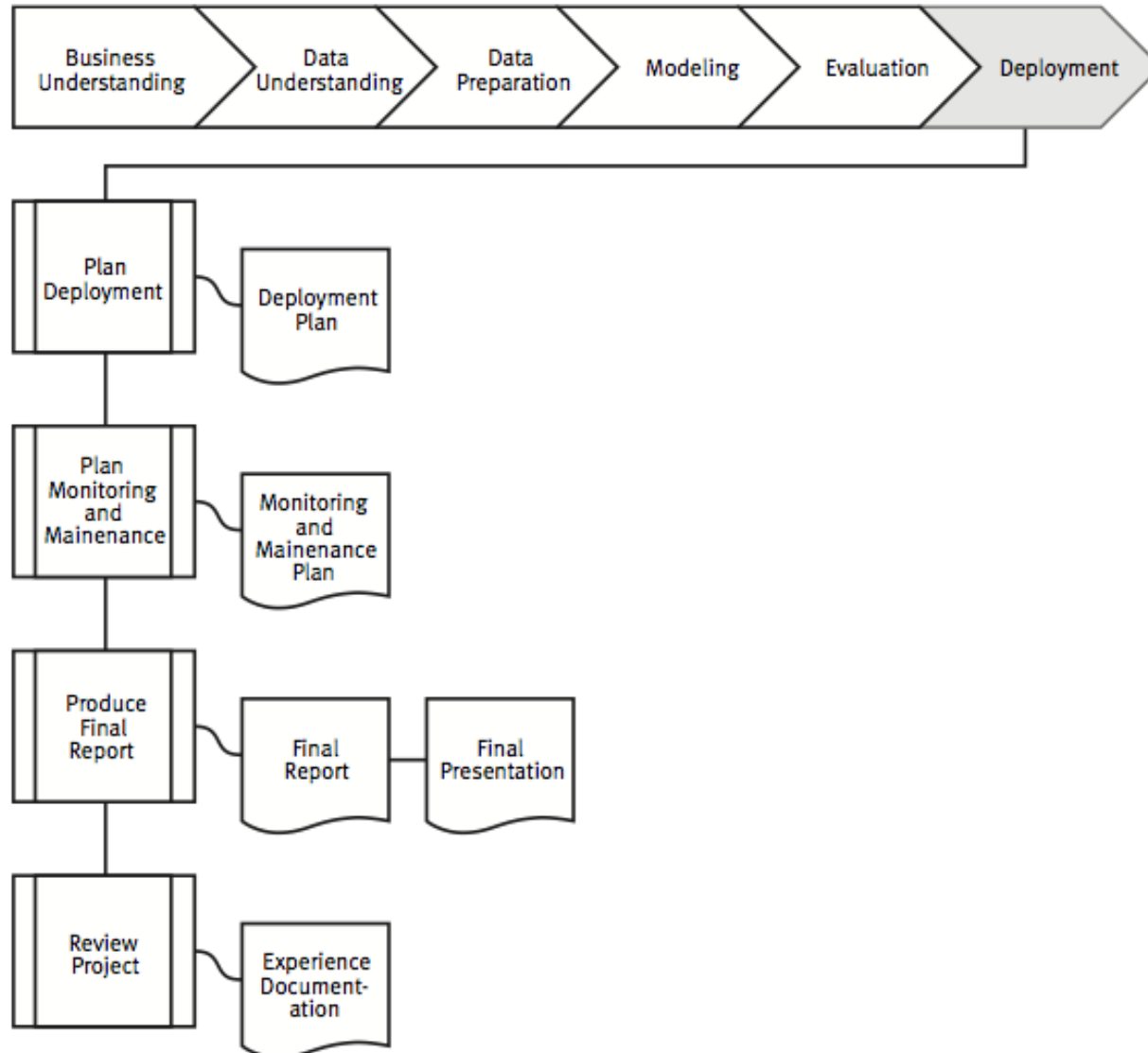
Review of project success

- Has the project achieved the original business objectives?
- Are there new business objectives to be addressed later in the projector in new projects?
- Conclusions for future datamining projects

Next Steps

- This section assesses the effectiveness of the project and identifies any factors that may have been overlooked that should be taken into consideration if the project is repeated
- List of possible actions

Deployment



Deployment plan

This report specifies the deployment of the data mining results

- Summary of deployable results (derived from Next Steps report)
- Description of deployment plan

Monitoring and maintenance plan

For each deployed result:

- Description of how updating will be triggered (regular updates, trigger event, performance monitoring)
- Description of how updating will be performed
- Summary of the results updating process

Final report

- Summary of business understanding: background, objectives, and success criteria
- Summary of data mining process
- Summary of data mining results
- Summary of results evaluation
- Summary of deployment and maintenance plans
- Cost/benefit analysis
- Conclusions for the business
- Conclusions for future data mining